

Graphical Analysis, Local and Global Optima, and Well-Posedness

Learning objectives

After studying this chapter, you should be able to

- use level curves to solve small two-variable problems graphically;
- distinguish local, strict local, global, and strict global optimizers; and
- use compactness and coercivity to prove the existence of global optimizers.

Optimization is not only the task of solving equations. Before an algorithm is useful, we should know what kind of solution is being sought and whether such a solution exists.

Two-dimensional pictures make these questions concrete: the feasible set shows where we may move, while the level curves show how the objective changes.

2.1 The graphical method

For a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and a height $c \in \mathbb{R}$, a *level curve* at height c is

$$L_c := \{\mathbf{x} \in \mathbb{R}^2 : f(\mathbf{x}) = c\}.$$

In a minimization problem, imagine moving the level curve toward smaller values until it first touches the feasible set. The touching point is a candidate optimizer. This method is rigorous only when the geometry and all boundary cases are checked, but it reveals active constraints and suggests the algebraic calculation.

Important note: geometry and algebra play different roles

The picture suggests which part of the boundary may be active. Algebra then verifies the candidate, compares competing boundary pieces, and checks whether the objective can escape to $-\infty$ or $+\infty$.

In higher-dimensional settings, the graphical method becomes inapplicable directly. But the collaborations between geometry and algebra remain; see later chapters for details.

Example 2.1 (Projection onto a polygonal feasible set). Consider

$$\begin{aligned} \min_{x_1, x_2} \quad & (x_1 - 4)^2 + (x_2 - 6)^2 \\ \text{subject to} \quad & x_1 \leq 4, \quad x_2 \leq 6, \quad 3x_1 + 2x_2 \leq 18, \\ & x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

Denote the objective by $f(x_1, x_2) := (x_1 - 4)^2 + (x_2 - 6)^2$. Its level curves are circles

centered at $(4, 6)$. The feasible point closest to that center lies on $3x_1 + 2x_2 = 18$ (see Figure 2.1 for an illustration). On this line, define

$$g(x_1) := f\left(x_1, 9 - \frac{3}{2}x_1\right) = (x_1 - 4)^2 + \left(3 - \frac{3}{2}x_1\right)^2.$$

The relevant segment has $2 \leq x_1 \leq 4$, and

$$g'(x_1) = \frac{13}{2}x_1 - 17.$$

Consequently, the point

$$\mathbf{x}^* := \begin{pmatrix} 34/13 \\ 66/13 \end{pmatrix}$$

satisfies $f(\mathbf{x}^*) = 468/169$. The other edges and vertices give larger values, so $\mathbf{x}^*$ is the global minimizer.

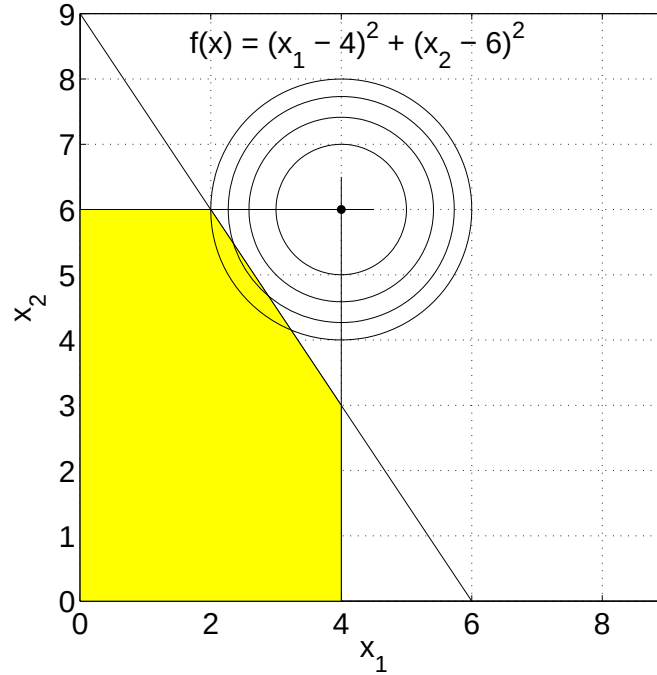


Figure 2.1: Circular level curves meeting a polygonal feasible region at a boundary optimizer.

Example 2.2 (An interior minimizer). Keep the same feasible set as in Example 2.1 but minimize

$$f(\mathbf{x}) := (x_1 - 2)^2 + (x_2 - 2)^2.$$

The center $(2, 2)$ of every circular level curve is feasible. Since $f \geq 0$ everywhere and $f(2, 2) = 0$, $(2, 2)$ is the global minimizer. No boundary calculation is needed.

Example 2.3 (A curved feasible boundary). Consider

$$\begin{aligned} \min \quad & x_1^2 - x_2 \\ \text{subject to} \quad & 2x_1 - x_2 \leq 4, \quad 9x_1^2 + 25x_2^2 \leq 225, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Write $f(x_1, x_2) := x_1^2 - x_2$ for the objective. For fixed x_1 , a smaller objective requires the largest feasible x_2 . The ellipse implies

$$x_2 \leq \frac{3}{5} \sqrt{25 - x_1^2} \leq 3,$$

whereas $x_1^2 \geq 0$. Thus $f(\mathbf{x}) \geq -3$, with equality at $(0, 3)$. Hence $(0, 3)$ is a global minimizer.

If the ellipse inequality is reversed, then $(0, t)$ is feasible for every $t \geq 3$, and $f(0, t) = -t \rightarrow -\infty$. The feasible set is then unbounded and the objective is unbounded below.

2.2 Local and global optimizers

Graphically, there may exist some feasible points that are not globally optimal but have objective values not worse (or even better) than the points in a small neighborhood. They may also be favorable to practitioners.

Definition 2.1 (Local and global minimizers). Let n be a positive integer, let $S \subseteq \mathbb{R}^n$, let $f : S \rightarrow \mathbb{R}$, and let $\mathbf{x}^* \in S$.

(i) $\mathbf{x}^*$ is a *local minimizer* if there is an $\varepsilon > 0$ such that

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \text{for every } \mathbf{x} \in S \cap B(\mathbf{x}^*, \varepsilon).$$

(ii) It is a *strict local minimizer* if the inequality is strict for every other feasible point in that neighborhood.

(iii) It is a *global minimizer* if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for every $\mathbf{x} \in S$.

(iv) It is a *strict global minimizer* if $f(\mathbf{x}^*) < f(\mathbf{x})$ for every $\mathbf{x} \in S \setminus \{\mathbf{x}^*\}$.

Local and global maximizers are defined by reversing the inequalities.

Every global minimizer is a local minimizer. The converse generally fails.

Example 2.4 (A local minimizer that is not global). Minimize x_2 subject to

$$x_2 \geq 10 - (x_1 - 3)(x_1 - 1)^2, \quad 0 \leq x_1 \leq 4.$$

At an optimizer, x_2 lies on the lower boundary. The boundary function has a local minimum at $x_1 = 1$, giving the local minimizer $(1, 10)$, but its value at $x_1 = 4$ is 1. Therefore $(4, 1)$ is a global minimizer and $(1, 10)$ is not.

Strictness concerns uniqueness near a point, not the size of the objective value. A strict local minimizer can coexist with a different point having a smaller objective value.

2.3 Well-posedness of optimization models

As we have emphasized before, checking the well-posedness of an optimization model should be the first step before solving it. For this purpose, we provide below two sufficient conditions. Readers should be careful that these conditions are not necessary and encouraged to raise counterexamples.

2.3.1 Continuous objective function on a compact set

Definition 2.2 (Bounded and compact sets). Let n be a positive integer. A set $S \subseteq \mathbb{R}^n$ is *bounded* if there is an $M > 0$ such that $\|\mathbf{x}\| \leq M$ for every $\mathbf{x} \in S$. In finite-dimensional Euclidean space, a set is *compact* if and only if it is closed and bounded.

Equivalently, $S \subseteq \mathbb{R}^n$ is bounded if and only if there is a $\widehat{M} > 0$ such that

$$|x_i| \leq \widehat{M} \quad \text{for every } \mathbf{x} \in S \text{ and } i = 1, \dots, n.$$

Indeed, $\|\mathbf{x}\| \leq M$ implies $|x_i| \leq M$ for every coordinate, whereas the coordinate bounds imply $\|\mathbf{x}\| \leq \sqrt{n} \widehat{M}$.

For a positive integer n , a point $\mathbf{x} \in \mathbb{R}^n$, and a radius $r > 0$, define the closed ball by

$$\overline{B}(\mathbf{x}, r) := \{\mathbf{y} \in \mathbb{R}^n : \|\mathbf{y} - \mathbf{x}\| \leq r\}.$$

Theorem 2.1 (Weierstrass extreme-value theorem). Let $S \subseteq \mathbb{R}^n$ be nonempty and compact, and let $f : S \rightarrow \mathbb{R}$ be continuous. Then f attains both a global minimum and a global maximum on S .

The theorem is an existence result. It does not identify the optimizer, and its assumptions are sufficient rather than necessary.

Example 2.5 (A compact feasible set). On the circle $x_1^2 + x_2^2 = 3$, the continuous function

$$f(x_1, x_2) := x_1^2 + 2x_2^2$$

attains both extrema by [Theorem 2.1](#). Since $x_1^2 = 3 - x_2^2$ on the circle,

$$f = 3 + x_2^2.$$

The minimum value is 3 at $(\pm\sqrt{3}, 0)$, and the maximum value is 6 at $(0, \pm\sqrt{3})$.

Example 2.6 (Existence without compactness). The line $x_1 + x_2 = 1$ is closed but unbounded, so [Theorem 2.1](#) does not apply directly. Nevertheless,

$$\min\{x_1^2 + 2x_2^2 : x_1 + x_2 = 1\}$$

has a solution. Substituting $x_2 = 1 - x_1$ gives $3x_1^2 - 4x_1 + 2$, whose unique minimizer is $(x_1, x_2) = (2/3, 1/3)$. Failure of a sufficient hypothesis is not proof of nonexistence.

Warning

Closedness, boundedness, continuity, and attainment are separate issues. For example, $(0, 1)$ is bounded but not closed; the continuous function $f(x) := x$ has infimum 0 there but no minimizer.

2.3.2 Coercive objectives on closed feasible sets

Definition 2.3 (Coercive function). A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is *coercive* if

$$\|\mathbf{x}\| \rightarrow \infty \implies f(\mathbf{x}) \rightarrow +\infty.$$

Equivalently, for every $M \in \mathbb{R}$ there is an $r > 0$ such that

$$\|\mathbf{x}\| > r \implies f(\mathbf{x}) > M.$$

Coercivity means that sufficiently distant points cannot compete with points in a bounded region. It is a growth condition. Some examples of coercive and non-coercive functions are shown in Figure 2.2.

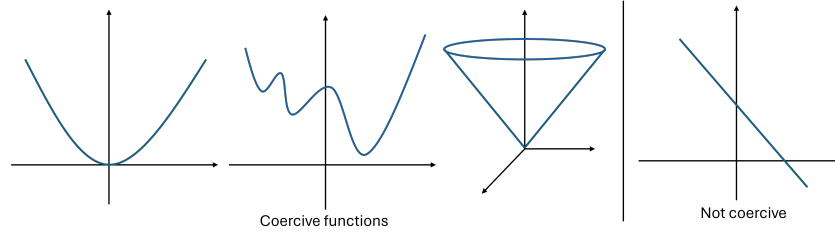


Figure 2.2: Coercive growth confines low objective values to a bounded region.

Example 2.7 (Coercive and noncoercive functions). The functions $x^4 + 3x^2 + 3$ on $\mathbb{R}$ and $x_1^2 + x_2^2$ on $\mathbb{R}^2$ are coercive. The function

$$f(x_1, x_2) := x_1^2 - 2x_1x_2 + x_2^2 = (x_1 - x_2)^2$$

is not coercive. Indeed, for $t \geq 0$, define $\mathbf{x}_t := (t, t)^\top$. Then $f(\mathbf{x}_t) = 0$ while $\|\mathbf{x}_t\| \rightarrow \infty$ as $t \rightarrow \infty$.

Example 2.8 (A polynomial with mixed signs). Let

$$f(x_1, x_2) := x_1^4 + x_2^4 - 3x_1^3 + x_2.$$

Define

$$\|\mathbf{x}\|_\infty := \max\{|x_1|, |x_2|\}, \quad r := \|\mathbf{x}\|_\infty.$$

Since $\|\mathbf{x}\| \rightarrow \infty$ if and only if $r \rightarrow \infty$,

$$\begin{aligned} f(\mathbf{x}) &\geq r^4 - 3r^3 - r \\ &= r^4 \left(1 - \frac{3}{r} - \frac{1}{r^3}\right) \geq \frac{3}{5}r^4 \quad (r \geq 10). \end{aligned}$$

Hence f is coercive.

Theorem 2.2 (Existence under coercivity). *Let $S \subseteq \mathbb{R}^n$ be nonempty and closed. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous and coercive, then f attains a global minimum on S .*

Proof. Choose $\mathbf{x}_0 \in S$ and set $M := |f(\mathbf{x}_0)| + 1$. Coercivity gives an $r_0 > 0$ such that

$$\|\mathbf{x}\| > r_0 \implies f(\mathbf{x}) > M > f(\mathbf{x}_0).$$

Define

$$r := \max\{r_0, \|\mathbf{x}_0\|\}, \quad K := S \cap \overline{B}(\mathbf{0}, r).$$

The set K is nonempty because it contains $\mathbf{x}_0$. It is also closed and bounded, hence compact. By Theorem 2.1, the restriction of f to K has a minimizer $\mathbf{x}^*$. Because $\mathbf{x}_0 \in K$,

$$f(\mathbf{x}^*) \leq f(\mathbf{x}_0).$$

For every $\mathbf{x} \in S \setminus K$, one has $\|\mathbf{x}\| > r \geq r_0$, and thus $f(\mathbf{x}) > f(\mathbf{x}_0) \geq f(\mathbf{x}^*)$. Therefore $\mathbf{x}^*$ minimizes f over S . $\square$

Example 2.9 (Revisiting Example 2.6). For the optimization problem

$$\min\{x_1^2 + 2x_2^2 : x_1 + x_2 = 1\},$$

define the feasible line by

$$S := \{\mathbf{x} \in \mathbb{R}^2 : x_1 + x_2 = 1\}.$$

The set S is nonempty and closed, while the objective is continuous and coercive. Hence Theorem 2.2 proves existence even though S is unbounded. The exact calculation in Example 2.6 identifies the unique minimizer.

Important note: a reusable existence strategy

Use growth to confine every serious candidate to a compact set, apply Weierstrass on that compact set, and finally show that points outside it have larger objective values.

Chapter summary

- Level curves show the direction of improvement; the feasible boundary determines where the last attainable level touches.
- Global optimality implies local optimality, but not conversely.
- Different local minimizers may have different objective values.
- A continuous function attains its extrema on a nonempty compact set.
- If $S \subseteq \mathbb{R}^n$ is nonempty and closed, then every continuous coercive function on $\mathbb{R}^n$ attains a global minimum over S .
- Compactness and coercivity are sufficient tools for existence, not necessary conditions.

2.4 Exercises

Exercise 2.1 (Sketching norm-defined sets). Sketch

$$S_1 := \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| < 4\}, \quad S_2 := S_1 \cap \{\mathbf{x} : 3x_1 - x_2 = 3\},$$

and

$$S_3 := \left\{ \mathbf{x} \in \mathbb{R}^2 : 1 \leq \left\| \mathbf{x} - (1, 2)^\top \right\| \leq 5 \right\}.$$

State which sets are open, closed, bounded, and compact.

Exercise 2.2 (A bounded union). Prove that the union of two bounded subsets of $\mathbb{R}^n$ is bounded. Does the same statement remain true for an infinite union? Justify your answer.

Exercise 2.3 (Graphical maximization over a strip). Solve graphically

$$\begin{aligned} \max \quad & x^2 + 2y^2 \\ \text{subject to} \quad & -1 \leq x + y \leq 1, \quad -1 \leq x - y \leq 1, \quad x \geq 0. \end{aligned}$$

Identify every global maximizer and the optimal value.

Exercise 2.4 (Contours on a triangle). Define $f(\mathbf{x}) := x_1^2/4 + x_2^2$. Consider

$$\begin{aligned} \max \quad & f(\mathbf{x}) \\ \text{subject to} \quad & x_1 + x_2 \leq 2, \quad x_1, x_2 \geq 0. \end{aligned}$$

- Sketch the feasible set.
- Sketch the level curves $f(\mathbf{x}) = 1$ and $f(\mathbf{x}) = 9/4$, and indicate the direction in which f increases.
- Determine all global maximizers.

Exercise 2.5 (A diamond-shaped feasible region). Solve graphically

$$\begin{aligned} \max \quad & \frac{x_1^2}{4} + (x_2 - 1)^2 \\ \text{subject to} \quad & -1 \leq x_1 + x_2 \leq 1, \quad -1 \leq x_1 - x_2 \leq 1. \end{aligned}$$

Draw the relevant level curves and identify every global maximizer.

Exercise 2.6 (Local versus global maximizers). Consider

$$\begin{aligned} \max \quad & x_1 + x_2 \\ \text{subject to} \quad & x_2 \leq x_1^2, \quad -2 \leq x_1 \leq 2. \end{aligned}$$

Sketch the feasible region, identify all local maximizers, determine whether a global maximizer exists, and decide whether every local maximizer is global.

Exercise 2.7 (Closedness, boundedness, and attainment). Let

$$D := \{\mathbf{x} \in \mathbb{R}^2 : -x_1^2 \leq x_2 \leq x_1^2, -1 \leq x_1 \leq 1\}.$$

- (i) Prove that D is closed.
- (ii) Prove that D is bounded.
- (iii) For $f(\mathbf{x}) := x_1^3 - 2x_2 + 8x_1x_2$, prove that f attains both a global minimum and a global maximum on D .

Exercise 2.8 (Three existence questions). For each problem, determine whether an optimal solution exists. Give a proof; finding the optimizer is not required unless it helps the argument.

(a)

$$\min \frac{x_1^4 + x_1^3x_2 + x_2^4 + 3}{x_1^4 + x_1x_2^3} \quad \text{subject to} \quad (x_1 - 2)^2 + x_2^2 \leq 1, \quad x_2 \geq 0.$$

(b)

$$\min x_1^2x_2 \quad \text{subject to} \quad x_1^2 - x_2^2 \leq 4, \quad x_2 \geq 0.$$

(c)

$$\min x_1 - x_2 \quad \text{subject to} \quad x_1^2 - x_2^2 = 4, \quad x_2 \geq 0.$$